

Weekly Progress Log - Week 2

Project Title: Data Science with Python

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Week: 2

Objectives for Week

- Understand relational databases and the basic structure of database tables.
- Learn how to retrieve and filter data using SQL queries.
- Practice core SQL commands including SELECT, WHERE, GROUP BY, and ORDER BY.
- Understand and practice JOIN operations across multiple related tables.
- Use aggregate functions such as COUNT, SUM, AVG, MIN, and MAX.
- Apply SQL to answer practical, business-style questions using a sample database.
- Become comfortable writing and testing SQL queries using SQLite or an online SQL practice platform.

Work Completed

- Reviewed the fundamentals of relational databases, including tables, rows, columns, primary keys, and relationships.
- Practiced basic SQL queries using SELECT to retrieve data from tables.
- Used WHERE clauses with comparison and logical operators to filter query results.
- Practiced sorting query results using ORDER BY in ascending and descending order.
- Used GROUP BY to organize data into groups for summary analysis.
- Practiced aggregate functions including COUNT, SUM, AVG, MIN, and MAX.
- Practiced INNER JOIN, LEFT JOIN, and RIGHT JOIN concepts using related tables.
- Completed SQL practice problems covering filtering, sorting, grouping, aggregation, and joins.
- Prepared a set of SQL queries that answer business-style questions on a sample database.

Challenges Faced

- Understanding how relational tables are connected using primary and foreign keys.
- Distinguishing between different JOIN types and determining which join is appropriate for a given question.
- Writing GROUP BY queries correctly when aggregate functions are used.
- Combining WHERE, GROUP BY, HAVING, and ORDER BY clauses in the correct SQL order.
- Interpreting business questions and translating them into accurate SQL queries.
- Handling NULL values and understanding how they affect filtering and aggregate results.
- Debugging syntax errors and checking query output for correctness.

Solutions Implemented

- Used small sample tables to visualize relationships between records before writing JOIN queries.
- Practiced each SQL clause separately before combining multiple clauses in a single query.
- Reviewed the logical order of SQL operations to write more accurate queries.
- Compared JOIN outputs to understand how INNER, LEFT, and RIGHT JOINS affect returned rows.
- Tested aggregate queries with different grouping columns to verify summary results.
- Checked query output against expected results and corrected syntax or logic errors when needed.
- Added comments to SQL scripts to explain the purpose of important queries.
- Organized SQL exercises and solved business questions into a dedicated Week 2 folder.

Individual Tasks

Responsibility	Description of Contribution	Remark
Relational Database Fundamentals	Reviewed tables, rows, columns, keys, and relationships in relational databases.	Built a foundation for writing SQL queries.
Basic SELECT Queries	Practiced SELECT statements to retrieve specific columns and records from sample tables.	Improved confidence retrieving data.
Filtering Data	Used WHERE with comparison and logical operators to filter records.	Practiced accurate data selection.
Sorting Results	Used ORDER BY to sort query results in ascending and descending order.	Improved result presentation and analysis.
Grouping Data	Used GROUP BY to organize records for summary analysis.	Learned how grouped reporting works.
Aggregate Functions	Practiced COUNT, SUM, AVG, MIN, and MAX on sample data.	Created summary statistics with SQL.
SQL Joins	Practiced INNER JOIN, LEFT JOIN, and RIGHT JOIN concepts across related tables.	Improved understanding of multi-table queries.
SQL Practice Problems	Solved SQL exercises involving filtering, sorting, grouping, aggregates, and joins.	Strengthened query writing and debugging skills.
Business-Style Queries	Prepared 10 or more SQL queries that answer practical questions using a sample database.	Completed the main Week 2 deliverable.
Repository Organization	Organized Week 2 SQL files and documentation for the DataScience_with_Python repository.	Week 2 work prepared for submission.